

Paarth Bhasin

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EDUCATION

Masters of Computer Engineering , <i>University of New South Wales - Distinction, WAM: 71.0</i>	2019 - 2022
Bachelor of Software Engineering , <i>Monash University</i>	2014 - 2018

SKILLS

Backend	Java, Python, Go, Node, S3, Protobuf, GraphQL, SQL, DynamoDB, MongoDB, NoSQL
Systems & Networking	Linux, Docker, Websockets, gRPC, Spark, Nginx, AWS, Redis, WireShark, Kubernetes
Frontend	React.js, Redux, JS/TypeScript, CSS, HTML, React Relay, Cypress, Jest, Bootstrap

EXPERIENCE

Senior Full Stack Engineer Big WX (https://bigw.com.au)	Nov 2024 - Present <i>Sydney, NSW</i>
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- E-commerce feature design and development, responsible for the order checkout, rewards/discounts, campaigns and loyalty card features on the website as well as the back-end. This is used by thousands of customers on a daily basis.
- Increasing test coverage of various user walkthrough scenarios in front-end.
- Working with **TypeScript**, **Kubernetes**, **Java**, **CSS/SCSS** & **Azure** to develop features related to digital customer rewards / loyalty cards. This involves improving business reporting requirements, critical bug-fixes as well as UX/UI improvements, and end to end testing improvements.

Senior Full Stack Engineer Evermed (https://evermed.com.au)	May 2024 - Oct 2024 <i>Sydney, NSW</i>
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- Rollout of new e-commerce web & back-end features, optimizing existing features, reducing tech-debt and improving operations in a fast-paced environment., with lots of pair-programming.
- Working with **AWS**, **Python**, **C#/.NET**, **CSS**, **Node.js**, **React.js**, **Nginx** for full-stack application development. Introduced test based approach cloud-infra and front-end applications, outlining testing strategy.

Senior Software Engineer Atlassian	Jan 2019 - Mar 2024 <i>Sydney, NSW</i>
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▷ Senior Platform Engineer - Jira Cloud Performance Atlassian (<i>Key achievements</i>)	Jan 2023 - Mar 2024 <i>Sydney, NSW</i>
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- Led initiative for performance optimization of a widely used user-facing endpoint serving millions of customers, by *algorithm redesign* & **SQL** query optimisation, improving endpoint latency by 50% on average, eg **91s → 35s**. This also involved filtering non-performant endpoints by using long term data storage via **AWS Kinesis** streams, to increase monitoring / observability of under-performant endpoints. This involved lots of collaboration and pair-programming across different geographies.
- Led improvement of database observability/capacity analysis - key metrics & optimising traffic/load distribution across multiple regions/data-centers.
- Drove **OpenTelemetry** adoption in Jira in **Java**, to adopt observability best-practices and for improving the tracing footprint. This involved evaluation of third-party tools as well such as **Grafana**, **NewRelic**, **Honeycomb.io**, **Prometheus**. This also involved *open source* contributions to OpenTelemetry's Java framework.
- Mentored junior engineers, to improve their working dynamics in a globally distributed team environment.
- Led an initiative to improve resiliency of database towards cluster fail-over, requiring OS level programming related with **JDBC** database driver and **TCP/IP** layer network optimization to reduce time to recover (TTR), for (**Aurora**) across the fleet in multiple regions/data-centers. This reduced TTR from 30-60m to 45-60s on average and <1s for best case scenarios.

- Devised testing framework for **performance analysis** of large customers in production coming to cloud, for multiple feature teams, playing key role in unblocking the migration of large customers (like Equifax - 50k+ users) to adopt our cloud services. This developed an underp-pinning strategy and principles for reliability, performance and scale testing strategy to conduct such testing for entire Jira Development organisation, in various geographies (~1500-2000 Developers).

▷ **Software Engineer - Enterprise Cloud**

Jan 2022 - Dec 2022

Atlassian (*Key achievements*)

Sydney, NSW

- Led the design and delivery of a large-scale runbook automation framework, using **Apache Airflow** (ETL Pipeline). This also involved **AWS Step Functions**, **AWS S3** & **AWS Lambda** to build automated workflows.
- Developed a UI, used for monitoring, tracing & management of deployments for synthetic tests, using **GraphQL** (using **Graphene** based **Python** GraphQL server), **React Relay** & **TypeScript**. It is used by 50+ engineering teams (>1000 developers), internally relying on a **Terraform** provider plugin, written in **Golang** & **Celery/SQS** (Python) based deployment service, that were feature-led by me. The plugin, allows a custom **Terraform** resource to be created and work with internal services, to allow **IaC** (Infrastructure as Code) for in-house developed tools.
- Improved Jira request tracing architecture, by incorporating tracing at the **Nginx** proxy level & changes on various critical request entry paths, ensuring consistency amongst all incoming requests & also did a *brown-bag*.

▷ **Software Engineer, Site Reliability Engineering**

Jan 2019 - Dec 2021

Atlassian (*Key achievements*)

Sydney, NSW

- Feature led improvement of async job framework (**Event Driven Architecture**) - introduced robust *health-checks* for Jira, including algorithm to detect false positives - by incorporating threshold for success & failure in **Java**, new key performance monitoring metrics for *scheduled asynchronous jobs*. Also delivered auto-scaling of instances based on discrete queue sizes.
- Developed automation (**Python**) for auto-scaling of **WebServer** (**AWS EC2**), in response to high tomcat thread loads in Jira.
- Managed all aspects of **CI/CD** pipelines - containerisation, deployments, and configuration as code in **Terraform**. This involved large scale scripting in **Terraform**, using best-practices like modules, remote state management, variables, version-management (**tfenv**), loops/counters.

Junior Software Engineer - Full Stack

June 2018 - Dec 2018

Odecee, A Cognizant Digital Business

Melbourne, VIC

- Full Stack Developer for development of an e-commerce website and back-end tests. Worked in an Agile Scrum team of 5, having fortnightly sprints and gained experience in Agile. Used **React.js**, **Redux** & **Spring MVC/Data**.

PROJECTS

• MapAI: Founding Engineer - Full Stack / Front End / Generative AI (<https://mapai.net/about>)

Apr 2024 - Jun 2024

Sydney, NSW

- Project funded by New South Wales Government, working as part of a university initiated 2-year research project.
- Full stack application design with best-practices for geo-spatial analysis tools working with Data Scientists, Researchers & Business Leaders. Building well designed user-interfaces to cater to the need of general audience. Working with **Python**, **Node.js**, **React.js**, **Deck.gl**, **Architectural / Design Patterns**, **Docker**, **Redis**, **Nginx**, **PostGIS** & **LLM** (Large language model) training using **OpenAI** & **LangChain** to provide geo-spatial searching, ranking & predictions.
- This also involved running the machine learning model on a personalised high powered GPU machine provisioned by the university, which was configured to be accessible by the internet through AWS VPC. public IP.

• LockR: Open-source Python Plugin for concurrency management

Jun 2022 - Dec 2022

- An open-source Python package, providing concurrency management of applications in the cloud with **Redis**, using *Redis Locking Pattern*. It can be accessed on [PyPI](#). It is used by multiple organisations and students.
- **Optiver Realized Volatility Prediction: Machine Learning Model** Jun 2021 - Aug 2021
 - A Machine Learning Model (LSTM), which helps with stock prediction. It can be accessed at: [Kaggle](#).
- **Amazon Web Services - Cyber Security Research Project: Unicorn** Sep 2019 - Dec 2019
 - A 3 month research project in the Amazon Web Services office, which focused on **continuous assurance** (CA) of applications deployed to AWS Cloud, AWS Cloud Infrastructure and Networks. This required building automation to implement validation of an application and its ecosystem (cloud infra).
- **Real-time 3D First Person Shooter - having Artificial Intelligence** Jun 2017
 - Developed a 3D first person shooter game having Artificial Intelligence (AI), using C++, Microsoft DirectX 11 and FMOD Sound Library.
 - Based on the famous Indie game Super Hot having a time warp/dilation effect.
- **Image Similarity Comparator - using Computer Vision** May 2017
 - A Python CLI App which calculates the percentage that a solid, partially filled image having noise, resembles/fits correctly in a hollow shape free of any noise. The hollow image and the solid image have same shapes, but can be of different sizes and orientation. The program takes care of that by resizing and de-skewing and also removes noise.
 - Uses ImageMagick and OpenCV to process and compare the images.
- **OpenGL: 3D Jet Plane Loader** Oct 2016
 - Loads data from a .dat file which represents a 3D Jet Model using **OpenGL** in **Java**. Follows **MVC** architectural pattern.

ACHIEVEMENTS & CERTIFICATIONS

- **Tutor Web Front-end Programming (COMP6080), UNSW - 2022** Imparted regular tutoring sessions to students. Additionally, my contributions to the course led to direct changes in the course framework such as Assignment criteria / updating lecture content, exam question assessments, by directly collaborating with the Lecturer.
- **Learn to Lead, UNSW - 2022** Learn to Lead with UNSW is a specially curated short course for UNSW alumni on the topic of leadership.